

- If the results are from the same distribution then the differences D_i should be symmetric about the point $\theta = 0 \rightarrow P(D_i \geq 0) = 1/2$ and $P(D_i \leq 0) = 1/2$.
 - If there is some bias in performance then differences D_i will be symmetric about the biased value $\theta = \theta^* \rightarrow P(D_i \geq \theta^*) = 1/2$ and $P(D_i \leq \theta^*) = 1/2$.
 - Most statistical texts describe the Wilcoxon signed ranks using a null hypothesis of $\theta^* = 0$ and alternative hypothesis of $\theta^* \neq 0$. This book customizes the hypothesis test for algorithmic comparison.
6. If performance across algorithms is equivalent then there is a 50% chance that $D_i > 0$ and a 50% chance that $D_i < 0$. The expected value and variance of our indicator function W is as follows:

$$E(W) = 1/2 \cdot 1 + 1/2 \cdot 0 = 1/2$$

$$V(W) = E(X^2) - [E(W)]^2 = 1/2 - (1/2)^2 = 1/4$$

7. This allows us to easily compute the expected value and variance of our summary statistic T_n . This is as follows:

$$E(T_n) = \sum_{i=1}^n i \cdot E(W_i) = \frac{1}{2} \cdot \sum_{i=1}^n i = \frac{n(n+1)}{4}$$

$$V(T_n) = \sum_{i=1}^n i^2 \cdot V(W_i) = \frac{1}{4} \cdot \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{24}$$

As $n \rightarrow \infty$, T_n converges to a normal distribution and we can use the standard normal distribution to determine our critical value:

$$Z_n = \frac{T_n - E(T_n)}{\sqrt{V(T_n)}}$$

Comparison to Critical Value:

- Reject the null hypothesis if $|Z_n| > C_{\alpha/2}$ where $C_{\alpha/2}$ is the critical value on the standard normal curve corresponding to the $1 - \alpha$ confidence level.
- For a 95% confidence test (e.g., $\alpha = 0.05$) we reject the null hypothesis if $|Z_n| > 1.96$.
- Above we are only testing if the distributions are different (therefore we use a two-tail test).
- This hypothesis can also be constructed to determine if A has better (or worse) performance than B based on whether $D_i > 0$ or $D_i < 0$ and using a one-tail test and corresponding critical values.